

THE RESEARCH JOURNEY

Final 13-Phase Documentary Shooting Bible

**Learning Socially Appropriate Group-Approach Behaviour for a TIAGo Robot
from Non-Expert Human Demonstrations**

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Shoot in reverse: 13 -> 1 | Edit in story order: 1 -> 13

Project: Learning Socially Appropriate Group-Approach Behaviour for a TIAGo Robot from Non-Expert Human Demonstrations

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Purpose: A complete phase-by-phase documentary record and reverse-shooting production guide, combining the 19 June meeting notes, project proposal, final dissertation, the uploaded Phase 4-13 technical documentary, repository history, exact project filenames, experiments, diversions, literature, AI-assistance notes, bugs, fixes, and recommended A-roll/B-roll/visual evidence.

Editing order: Phase 1 -> Phase 13.

Physical shooting order: Phase 13 -> Phase 1, progressively trimming the beard so the edited film appears to have been recorded chronologically.

Important dating rule

The first three phases can be anchored cleanly to the 19 June meeting and early July Git history. The late-August research became highly parallel: Gazebo world development, integration, detector work, fault diagnosis, V2 experiments, final trials and dissertation writing overlap. Therefore every phase below has a **documentary anchor date** and an **evidence window**. They are tracking anchors, not a claim that only one task happened on one day.

Evidence base

- Meeting (19-06-26).pdf - initial research objective and exploratory literature list.
- CMP9140_Assessment1_ProjectProposal_SaaranathanThuviprakash(30276855).pdf - proposal, objectives, ethics, Gantt chart and risks.
- CMP9140_Research-Project_Saaranathan_(30276855).pdf - 93-page final dissertation, methodology, results, 38-reference bibliography.
- PROJECT_DOCUMENTARY_PHASES_4_TO_13.md - detailed technical documentary record supplied for Phases 4-13.
- Public repository saaruba/Research_Project - commit history, exact file names, results folders, demo media and technical documentation.

MASTER TIMELINE

Phase	Documentary anchor	Evidence window	Main narrative
1	19 Jun	19-25 Jun	Supervisor meeting, broad problem, first literature search
2	26 Jun	20 Jun-2 Jul	Proposal, objectives, ethics, risks, measurable targets
3	3 Jul	3-10 Jul	TIAGo, Docker, ROS 1/ROS 2, tutorials and first repo commits
4	7 Aug	11 Jul-7 Aug	Dataset forensics, bag extraction, synchronisation, split
5	8 Aug	7-8 Aug	People -> groups -> O-space approximation -> BC labels
6	8 Aug	8-9 Aug	RF/MLP/rule/naive models, 56-config tuning, negative offline result
7	9 Aug	9-20 Aug	Full restaurant Gazebo world, actors, maps, collisions, ground truth
8	19 Aug	19-23 Aug	Closed-loop ROS 2 integration and reliability fixes
9	13 Aug	13-24 Aug	YOLO vs LocateAnything, Report 1 feedback, decoder/inference fixes
10	21 Aug	20-23 Aug	Localisation/collision/task-scoring measurement crisis
11	24 Aug	24 Aug	V2 segmentation, data sweeps, label ambiguity, 360-model search
12	25 Aug	25 Aug	Objective 3 hand-labelling/validation + writing/documentation push
13	24-25 Aug	24-30 Aug	Final experiment, slot metric, inversion, hybrid conclusion, submission wrap-up

HOW TO USE EACH PHASE

Every phase contains the story, literature, tools/repositories, failures/diversions, fixes, exact files to show, visual/actor guidance, AI-assistance note where relevant, and the handoff that causally creates the next phase.

PHASE 1 - THE BEGINNING

"What Have I Actually Signed Up For?"

Date / work window

Documentary anchor: 19 June 2026 **Evidence window:** 19-25 June 2026 **Dating basis:** Supervisor meeting record dated 19-06-2026.

The story

The research journey begins with the supervisor meeting recorded on 19 June 2026. The first working objective was broad: develop and evaluate a TIAGo system that learns from human demonstrations to approach small conversational groups at a socially appropriate position and angle while maintaining safe indoor navigation.

At this point the project was deliberately much wider than the final dissertation. The meeting notes list fourteen possible technical directions: human detection, group detection, F-formation recognition, human tracking, proxemics, approach-point selection, approach-angle selection, collision-free navigation, Learning from Demonstration, Behavioural Cloning, gaze-aware decision making, language-aware decision making, multimodal learning and social-compliance prediction. This is important for the documentary because Phase 1 should not pretend the final architecture was already known.

The early perception idea was also ambitious: use LocateAnything-3B on an external NVIDIA GPU machine so TIAGo could answer "Where are the people?" and "What objects are around them?", while F-formation, proxemics and Behavioural Cloning would answer "Are they a group?", "Where should TIAGo stand?" and "From which angle should it approach?"

The papers and research trails

The 19 June meeting record already contains a large exploratory reading list. The most important early papers include Faris et al. on PLUS-HRI non-expert demonstrations, Kruse et al. on human-aware robot navigation, Singamaneni et al. on socially aware navigation, Argall et al. on Learning from Demonstration, Setti et al. on F-formations, Yang et al. on human/robot group approach behaviour, Gao et al. on learning group-approach behaviour with deep reinforcement learning, and related work on robot positioning, cultural differences, F-formation perception, dynamic proxemics, gaze and socially compliant navigation.

The point of showing these papers in Phase 1 is not to explain all of them. Show the viewer that the first week is an exploration problem: there are many valid ways to frame social approach, and the project has not yet decided which one is feasible.

Tools, repositories and first technical exploration

- PLUS-HRI project/dataset material.
- PAL Robotics TIAGo platform and simulation resources.
- ROS / Gazebo / RViz ecosystem.
- LocateAnything-3B paper and model resources.
- GitHub and a new project repository that will later become Research_Project.

ChatGPT (Sara) involvement

Saarunathan led the project and supervisor discussion. ChatGPT ("Sara") assisted during this early period with literature-search keywords, paper/repository exploration, explaining social-robotics concepts, organising the project pipeline and turning the broad meeting ideas into a researchable sequence. The supervisor remained the academic decision-maker and Saarunathan made the final project choices.

What to film / show

A-roll: clean-shaven/groomed beginning-of-project diary. Speak as someone who has more questions than answers.

B-roll / screen:

- Meeting (19-06-26).pdf, especially the first page with the core objective and the 25-paper exploratory list.
- Search tabs for Kruse, Argall, Setti, Yang, Gao and LocateAnything.
- A blank notebook page becoming a rough pipeline: People -> Groups -> Where to stand -> Nav2.
- Campus / lab exterior; a shot of TIAGo imagery or the PAL Robotics webpage.

Presenter direction: do not explain Behavioural Cloning or O-space like an expert yet. The emotional beat is excitement plus uncertainty.

The handoff

"I have a huge list of things I could do. Now I need to turn it into one research question I can actually finish and measure."

PHASE 2 - THE PROPOSAL

"Turning a Huge Idea Into Something I Can Actually Test"

Date / work window

Documentary anchor: 26 June 2026 **Evidence window:** 20 June-2 July 2026 **Dating basis:** Proposal drafting/feedback window; project Gantt begins week of 29 June.

The story

The broad Phase 1 concept is narrowed into the formal dissertation project: learn the final stopping position and orientation of a robot approaching a small conversational group from an existing anonymised non-expert demonstration dataset, and compare that learned behaviour with a hand-engineered geometric rule in simulation.

The proposal defines four objectives: review the literature; implement and validate person perception; detect groups and estimate O-space; and train/compare Behavioural Cloning against a rule-based baseline. It also defines a simulation-only scope, no new participants, no physical TIAGo deployment, and a quantitative evaluation plan.

The proposal is where the project gains its first measurable promises. It also records the ethics/LEAS reasoning: the work uses an existing anonymised dataset and simulation, and after discussion with the supervisor a new LEAS application was not required for the defined scope. If the scope later changed to real TIAGo testing or participant feedback, fresh ethics approval would be required.

The theory that becomes the foundation

- Hall (1966), proxemics: interpersonal distance carries social meaning.
- Kendon (1990), F-formations: conversational groups organise around an O-space, with participants in the P-space and a newcomer joining through a suitable opening.
- Kruse et al. and Rios-Martinez et al.: human-aware/socially aware navigation is more than collision avoidance.
- Setti et al.: computational F-formation detection.
- Argall et al.: Learning from Demonstration and Behavioural Cloning.
- Gao et al.: a learned group-approach alternative using reinforcement learning.
- Yang et al.: one of the closest datasets/studies of human and robot approaches to small conversational groups.
- Faris et al./PLUS-HRI: useful behaviour can be learned from imperfect demonstrations by ordinary users.

First supervisor-feedback loop

The proposal-feedback round is distinct from the later Report 1 detector feedback. This early feedback strengthened missing citations, explicit limitations in the literature review, measurable success criteria and the Behavioural Cloning methodology. The project locked in the main numerical success targets used later in the dissertation: person detection recall $\geq 80\%$; O-space validation $\geq 70\%$; approach position error < 0.4 m; approach orientation error < 20 deg.

The proposal risk table becomes a documentary foreshadowing device. It predicts missing dataset information, LocateAnything limitations, group/F-formation errors, Behavioural Cloning generalisation problems, unreachable goals, excessive scope, and the limitation of using simulation proxy metrics instead of real human comfort. Many of those risks later happen.

Scope reduction / diversions

Gaze-aware reasoning, language-aware behaviour and multimodal learning are not forgotten; they are deliberately moved out of the core deliverable so perception, grouping, learning, navigation and evaluation can be completed within an MSc project.

ChatGPT (Sara) involvement

ChatGPT assisted Saarunathan in refining the early idea into a clearer proposal narrative, explaining and organising proxemics/F-formation/LfD concepts, discussing objectives and evaluation, reviewing literature directions, and helping plan the project workflow. The final proposal content, academic submission and supervisor decisions remained Saarunathan's responsibility.

What to film / show

A-roll: short beard/very groomed. "This week the idea stopped being a cool robot idea and became a list of things I will be judged on."

B-roll / source pages:

- CMP9140_Assessment1_ProjectProposal_SaarunathanThuviprakash(30276855).pdf.
- Proposal p.3: aim, research question and four objectives.
- Proposal pp.5-6: methods and ethics.
- Proposal p.7: Gantt/project plan - use as the "what I thought would happen" shot.

- Proposal p.8: risk table - use as a recurring visual later; tick risks as they materialise.
- Proposal_Feedback_Action_Plan.md for the first feedback round.

On-screen graphic: Aim -> 4 Objectives -> 4 numeric targets.

The handoff

"The proposal tells me what the robot should do. Now I have to learn the TIAGo and ROS stack well enough to build it."

PHASE 3 - MEETING TIAGO

"The Proposal Looked Much Easier on Paper"

Date / work window

Documentary anchor: 3 July 2026 **Evidence window:** 3-10 July 2026 **Dating basis:** GitHub first push 3 July; TIAGo Docker test 5 July; head-control/navigation update 10 July.

The story

The project enters its first hands-on robotics stage. The GitHub repository receives its first push on 3 July 2026. On 5 July the record explicitly shows TIAGo being installed/tested in Docker and played with in simulation, and on 10 July head-control and navigation updates are committed.

The central learning problem is the ROS ecosystem. The source demonstrations come from ROS 1 recordings, while the implementation is moving toward Ubuntu 22.04 and ROS 2 Humble. TIAGo tutorials, launch files, topics, TF frames, Gazebo and RViz have to be understood before the social-approach research can even begin.

This is also where the practical difference between "a research idea" and "a robot system" becomes obvious. A paper can say "navigate the robot to a pose" in one sentence. In practice there are packages, controllers, transforms, QoS profiles, navigation stacks and simulation dependencies underneath that sentence.

Literature and documentation

- Pages, Marchionni and Ferro (2016) on the TIAGo platform.
- PAL Robotics TIAGo tutorials/documentation and simulation repositories.
- ROS 2 Humble documentation.
- Nav2 documentation begins to become relevant, although the full closed-loop use comes later.

Repositories and files to show

- GitHub commit ca30350 - "First Push" (3 July 2026).
- Commit a84bcb7 - TIAGo in Docker/testing in simulation (5 July 2026).
- Commit e7992c3 - TIAGo head control/navigation updates (10 July 2026).
- docs/tiago_setup_progress.md.
- src/tiago_head_control/ - early head-aiming/debugging package.

What went wrong / learning diversions

The key "failure" in this phase is not one bug; it is the learning curve. Tutorials and packages built around different ROS generations do not line up automatically. Environment setup, Docker, dependencies and simulation launch behaviour take time. Some early work - for example head control - does not become the dissertation's headline contribution, but it is essential scaffolding and remains in the final repository as a debugging aid.

ChatGPT (Sara) involvement

Saarunathan used ChatGPT extensively during the TIAGo-learning stage for ROS/TIAGo explanations, Docker/ROS version troubleshooting, tutorial interpretation, package organisation and command-level guidance. The user performed the actual installs, builds and simulation tests.

What to film / show

A-roll: still early-project energy, but the first "this is harder than the proposal" moment.

B-roll:

- Terminal launching TIAGo in Docker.
- Gazebo/RViz with TIAGo visible.
- A ROS topic list and TF tree.
- GitHub commits dated 3, 5 and 10 July.
- tiago_head_control code or the robot head moving.

Optional visual explainer: split screen: ROS 1 bag icon on left, ROS 2 Humble/TIAGo on right, with "data world" and "deployment world" labels.

The handoff

"I can finally run and control TIAGo. The next question is what is actually inside the demonstrations it is supposed to learn from."

PHASE 4 - DATASET FORENSICS

"What Is Actually Inside These Bag Files?"

Date / work window

Documentary anchor: 7 August 2026 **Evidence window:** 11 July-7 August 2026 **Dating basis:** Processed research data and ML outputs were committed on 7 August after the July setup period.

Note: Later project phases overlap because experiments, debugging, writing and analysis happened in parallel. These dates are tracking anchors, not claims that work stopped at phase boundaries.

The story

Up to this point the dataset had been an abstraction - "the recordings my supervisor gave me." Phase 4 is where it becomes 24 folders of ROS 1 bag files, videos, CSVs and timestamps that do not agree with each other.

PLUS-HRI consists of **24 recorded non-expert teleoperation sessions**. Ordinary people, not roboticists, drove a robot around and approached groups. The premise of the whole project is that their choices contain useful judgement about where it is socially appropriate to stand.

Direct inspection - rather than assuming a uniform format - established something that shaped everything afterward: **the dataset spans two distinct recording generations**.

- **Sessions 1 and 3** carry full per-frame facial-landmark annotations (`facial_landmarks_uniface.csv`) from an earlier pipeline.
- **The other 22 sessions** were recorded later in a segmented-clip format carrying only `cmd_vel.csv`, the operator's own raw eye-gaze coordinates (the *driver's* eyes, not the people being approached), the robot's head pan/tilt joint state, and joystick input - **with no person-position ground truth of any kind**.

Then the finding that redirected the entire methodology: `gaze_uniface.csv` was confirmed by direct inspection to be **entirely empty - 100% null across every column, in every session**.

There is no per-person facing-direction ground truth anywhere in this corpus.

That single fact removes the possibility of doing classical F-formation detection the way the literature does it, because the standard methods need to know which way people are facing. Phase 5 exists because of this discovery.

Extraction (`extract_training_table.py`) read each session's ROS 1 bags through the `rosbags` library's ROS 1 reader and `typestore`, and merged pose (`/robot_pose`, `/mobile_base_controller/odom`, `/dlo_node/odom`), LiDAR (`/scan`) and person-detection topics onto a common timestamp index with `pandas.merge_asof`. Across all 24 sessions this produced **229,678 synchronised rows at approximately 40 Hz**.

[AI assistance note: The initial extraction pipeline was developed by Saarunathan with ChatGPT ("Sara") assistance during the earlier part of the project. The later documentary record reconstructs the technical details from the code, methodology and dataset.]

The papers

Reference	Why it belongs here
Argall, Chernova, Veloso & Browning (2009) - <i>A Survey of Robot Learning from Demonstration</i>	Stops being abstract. You are now physically looking at what a "demonstration" consists of: state, action, timestamp
Ravichandar et al. - <i>Recent Advances in Robot Learning from Demonstration</i>	Demonstration <i>quality</i> and <i>coverage</i> , not quantity, is what determines what can be learned
Faris, Kucukyilmaz, Polydoros & Del Duchetto (2025) - PLUS-HRI recovery policies	The precedent that imperfect non-expert demonstrations still contain usable behavioural signal

The tools

`rosbags` (ROS 1 reader + `typestore`) · `pandas` (`merge_asof`) · `NumPy` · `OpenCV` · the 24-session PLUS-HRI corpus (~70 GB raw, deliberately excluded from git)

What went wrong

1. Corrupt green video frames. `cv2.CAP_PROP_POS_FRAMES`-based seeking decoded solid-green images at certain offsets. Silently. A frame that looks like a rendering glitch produces detections of nothing, and those propagate into training labels.

2. 80-88% of rows dropped to missing data. The initial merge tolerance between `cmd_vel` and detection timestamps was 0.1 seconds. But the 22 segmented sessions sample detections at roughly **1 Hz**, not per frame. A tenth-of-a-second window almost never finds a match.

3. Every detection matching every frame. `np.isclose` defaults to a *relative* tolerance of `rtol=1e-5`. On a Unix timestamp of about 1.76×10^9 , that relative tolerance evaluates to roughly **17,000 seconds** - nearly five hours. Every timestamp comparison returned true.

That third one is the most instructive bug in the whole project. It produces no error, no warning, and no obviously wrong output. It just quietly makes the data meaningless.

How it was tackled

1. `extract_person_detections.py` rewritten to read frames **strictly sequentially** rather than seeking. Slower, correct.
2. Merge tolerance widened to **0.6 s** - half the ~1 Hz sampling gap on either side. Critically, this does not change *which* detection `merge_asof` selects as nearest; it only stops valid matches being discarded.
3. `rtol=0` passed explicitly wherever timestamps are compared.

And the split. `split_dataset.py` allocates **whole sessions** to train/validation/test (70/15/rest, fixed seed 42) so no session's rows appear in two splits. Rows within a session are heavily autocorrelated - near-identical pose and LiDAR a few tenths of a second apart - so a row-level split would let a model memorise a session instead of generalising.

Sessions 1 and 3 are **forced into training** rather than randomly assigned, because they are the only two with real human-position ground truth and too valuable to risk holding out. This is declared as a limitation in the dissertation, not hidden: the perception and O-space validations therefore validate against sessions the model also trained on. That is acceptable for validating a fixed geometric pipeline, and unacceptable for validating a learned model - which is why the BC evaluation uses fully held-out sessions **5, 9 and 59 (11,921 rows)** that were never touched by any training or tuning decision at any point, including the later V2 study.

Final split: 17 train (44,190 rows) / 4 validation (14,444) / 3 test (11,921).

What to film / show

A-roll: explain the first shock: the dataset is not one clean table; it is 24 sessions with different recording generations.

Exact files/code on screen: `scripts/extract_training_table.py`, `scripts/extract_person_detections.py`, `scripts/inspect_dataset_session.py`, `scripts/split_dataset.py`, `dataset/processed/ outputs`.

Source pages/images: Dissertation pp.31-33 (dataset and split table). Show a folder tree containing `.bag`, video and CSV files. Use an animated timestamp line to explain `merge_asof` and the `rtol=0` bug.

Visual gag that is technically useful: flash a solid-green corrupted frame and caption it "valid-looking pipeline, invalid frame". Then show 0.1 s -> 0.6 s tolerance and the 17,000-second `np.isclose` trap.

AI-credit line if spoken: "I used ChatGPT heavily in the early extraction/data-understanding stage, but I ran the dataset inspection and made the final preprocessing decisions."

The handoff

"I have robot logs, images and commands. But none of that is a learning problem yet. How do I turn a room full of people into groups, and a driving session into labelled examples?"

PHASE 5 - BUILDING THE SOCIAL REPRESENTATION

"People Become Groups, Groups Become Training Data"

Date / work window

Documentary anchor: 8 August 2026 **Evidence window:** 7-8 August 2026 **Dating basis:** Approach-pose learning/evaluation pipeline committed 8 August.

Note: Later project phases overlap because experiments, debugging, writing and analysis happened in parallel. These dates are tracking anchors, not claims that work stopped at phase boundaries.

The story

This is where the social-science theory becomes code.

Grouping. `cluster_groups.py` treats individually detected people, per frame, as nodes in a similarity graph. Two people join the same group when their pixel distance falls below a threshold - but expressed in **units of their own average bounding-box width, not raw pixels**. Connected components become candidate groups.

That normalisation is a deliberate, declared approximation for a real constraint: the recorded video has no camera calibration and no depth. Bounding-box width shrinks with distance from the camera in roughly the same way true separation does, so measuring in "person-widths" partially cancels the perspective distortion that raw pixel distance would not.

O-space. Kendon's F-formation framework describes the shared, jointly-oriented space at a conversational group's centre. The framework's usual construction needs orientation - which Phase 4 proved does not exist in this dataset.

The adopted fallback is the **mutual-facing assumption**: the O-space centre is approximated as the group centroid produced by clustering, and (in the live system) the O-space radius as the mean distance from that centroid to its members.

This is not an ad-hoc workaround, and the documentary should be clear about that. It is independently justified by **Vascon et al. (2016)**, whose game-theoretic F-formation detection demonstrates that group structure can be recovered **from relative position alone**, without orientation estimates. The project leans on that result directly to defend the simplification as legitimate rather than merely convenient.

Labels. No stage of the raw recording says "this was a demonstrated group approach." It had to be inferred. `build_approach_pose_dataset.py` finds transitions from sustained movement to a genuine stop with a detected group nearby; that stop pose becomes the operator's implicit judgement of "a good place to be near this group," and every row leading up to it is labelled with that eventual stop pose as its target.

Result: **462 independent demonstrated approach events across 24 sessions, expanding to 70,555 labelled rows.**

The papers

Reference	Why it belongs here
Kendon (1990) - <i>Conducting Interaction</i>	The origin of F-formations, O-space and P-space. The theoretical spine of the entire project
Hall (1966) - <i>The Hidden Dimension</i>	Proxemics. The 0.45-1.2 m personal zone that later becomes the rule policy's 1.2 m standoff
Setti, Russell, Bassetti & Cristani (2015) - F-formation detection in images	The canonical computational treatment - and the one that needs orientation
Cristani et al. (2011) - Social interaction discovery by statistical analysis of F-formations	The statistical lineage of the same problem
Vascon et al. (2016) - game-theoretic F-formation detection	The load-bearing citation. Position alone is sufficient. This is what makes the fallback defensible rather than a compromise
Swofford et al. - DANTE	The learning-based alternative, considered and rejected: this corpus has no pairwise conversational-membership labels to train it

The tools

OpenCV · NumPy · pandas · scikit-learn · `cluster_groups.py`, `build_approach_pose_dataset.py`, `estimate_approach_points.py`

What went wrong

``cmd_vel`` in this dataset is **extremely spiky**. Isolated single-sample non-zero values rather than sustained speed changes. A naive sample-by-sample moving/stopped test found almost no usable segments at all - the segmentation simply failed.

Absolute position made the model worse. An early baseline included raw robot (x, y) as a feature. It measured *worse* than omitting it, because absolute position does not generalise across sessions recorded in different rooms.

How it was tackled

The spiky velocity was handled by averaging speed into **0.5-second bins** before applying the moving/stopped test. That recovered genuine, usable movement segments.

The position problem produced a design decision that shaped everything downstream: the prediction target is the stop pose expressed **relative to the robot's own position and heading at prediction time** - Δx , Δy in the robot's forward/left frame plus Δyaw - rather than an absolute map coordinate. If absolute position doesn't generalise as an input, it certainly won't as an output.

The seven features, in a fixed order the deployed policy node must reproduce exactly:

Feature	Meaning
lidar_min_range	Distance to the nearest obstacle
lidar_mean_range	Average distance all round - how open the space is
linear_x_prev	How fast the robot was just driving
angular_z_prev	How fast it was just turning
num_people	How many people are visible
group_bearing_rad	Which direction the group is in
group_scale_norm	How large they appear - a proxy for closeness

Targets: target_dx, target_dy, target_dyaw.

Note what is *not* in that list: no images, no absolute coordinates, no orientation of any person. Seven numbers.

What to film / show

A-roll: draw three people as circles, an O-space in the middle, and show why missing facing direction forces a positional approximation.

Exact files/code: scripts/cluster_groups.py, scripts/build_approach_pose_dataset.py, scripts/estimate_approach_points.py.

Source pages/images: Dissertation pp.36-40. Put the seven-feature table on screen and animate the three targets target_dx, target_dy, target_dyaw from the robot frame.

Paper visual: Kendon -> Setti/Cristani -> Vascon. Make Vascon the key "this is why position-only is defensible" moment.

The handoff

"I finally have situation -> demonstrated stopping pose. So: can a model actually learn it?"

PHASE 6 - TEACHING THE ROBOT

"Behavioural Cloning vs Geometry"

Date / work window

Documentary anchor: 8 August 2026 **Evidence window:** 8-9 August 2026 **Dating basis:** Initial learned-model training/evaluation and environment pipeline consolidated in the 8 August build-out.

Note: Later project phases overlap because experiments, debugging, writing and analysis happened in parallel. These dates are tracking anchors, not claims that work stopped at phase boundaries.

The story

Four policies, trained and compared honestly:

1. **Naive mean predictor** - ignores every input, always predicts the training-set average. The floor.
2. **Rule-based geometric baseline** - F-formation and proxemic geometry, applied offline. The control condition.
3. **Random Forest** - a lower-variance ensemble.
4. **MLP** - the proposal's committed primary architecture.

The rule baseline being the control condition matters scientifically. The project is not trying to prove "AI is better." It is asking whether learning adds anything over an interpretable geometric solution that a person could write down.

Both learned families were tuned by grid search (`grid_search_approach_pose.py`) exactly as the proposal committed: **every candidate scored only on the validation split**, with the test split held out entirely until one winner per family was selected and evaluated once.

Search size: 40 Random Forest configurations + 16 MLP configurations = 56.

Tuning materially helped:

Model	Before tuning	After tuning
Random Forest	0.401 m / 29.3°	0.365 m / 25.8°
MLP	0.466 m / 42.2°	0.395 m / 31.0°

The papers

Reference	Why it belongs here
Pomerleau - ALVINN	The origin of behavioural cloning for vehicles. A neural network learning to steer by watching a human
Bojarski et al. (2016) - End to end learning for self-driving cars	The modern restatement of the same idea
Argall et al. (2009)	The LfD framing: fit the policy to human choices rather than engineering a rule
Breiman (2001) - Random forests	The justification for the ensemble alternative
Gao et al. (2019) - deep RL for group approach	The road not taken. RL needs a designed reward; BC learns from a fixed corpus, which is what exists here
Tai et al. (2018) - GAIL from raw depth	Another explored-and-not-chosen imitation approach

The tools

scikit-learn · joblib · `grid_search_approach_pose.py`, `evaluate_approach_pose.py`

What went wrong

The proposed architecture was too big. The winning MLP configuration was **32/16 hidden units with L2 regularisation $\alpha = 0.1$** - substantially smaller and more heavily regularised than the 128/64 network the proposal committed to. Not merely under-tuned. Too large for the amount of independent demonstration data available.

And then the result nobody wants. The full held-out evaluation on sessions 5, 9 and 59:

Policy	Mean pos. (m)	Median pos. (m)	Mean orient. (°)	Within position	Within orientation	Within both
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Naive (predict mean)	0.410	0.365	28.97	68.4%	63.2%	47.2%
Rule-based (geometric)	0.305	0.164	29.13	70.1%	56.8%	43.1%
Random Forest (untuned)	0.401	0.294	29.27	61.4%	55.8%	38.2%
MLP (untuned)	0.466	0.353	42.22	55.3%	37.5%	21.8%
Random Forest (tuned)	0.365	0.267	25.78	66.8%	61.3%	43.3%
MLP (tuned)	0.395	0.305	30.99	61.9%	47.0%	31.8%

No policy met the 20° orientation threshold. And the naive mean predictor's 47.2% beats every learned model on the joint criterion.

A model that ignores all seven inputs and predicts the average is competitive with everything you just built.

How it was tackled

Not by explaining it away. By **testing whether it was an artefact of one grid search**. A separate sensitivity check on a *different* random session split, six configurations across both families:

Configuration	Test position (m)	Test orientation (°)
Naive (predict training mean)	0.349	25.1
Random Forest, unregularised	0.351	30.4
Random Forest, leaf = 5, depth = 20	0.341	28.9
Random Forest, leaf = 20, depth = 15	0.331	27.1
MLP 128-64	0.507	38.0
MLP 64-32, $\alpha = 1e-2$	0.401	34.6

Same pattern, independent split. The naive predictor beats every learned model on orientation.

At this point the suspicion was recorded honestly: the limiting factor is probably **demonstration volume, not model capacity**. That suspicion sat unresolved for a month. Phase 11 is where it gets tested properly.

What to film / show

A-roll: "For the first time I can press Train and get numbers back - and the numbers are not flattering."

Exact files/code: scripts/grid_search_approach_pose.py, scripts/evaluate_approach_pose.py, scripts/train_approach_pose_model.py, scripts/train_approach_pose_mlp.py, scripts/train_baseline_model.py, scripts/train_social_baseline_model.py.

Source pages/images: Dissertation pp.39 and 62-63 for tuning method; p.68 for the offline-results table. Show the 56-config search counter and a simple bar/number reveal of tuned RF and MLP.

Actor beat: when the naive mean predictor beats the learned models on the joint criterion, stop the montage and address the camera. This is the first real negative result.

The handoff

"A regression error in a Python table doesn't tell me whether a robot behaves socially in a room. I need somewhere to run this."

PHASE 7 - BUILDING THE RESTAURANT

"Somehow My AI Dissertation Became Level Design"

Date / work window

Documentary anchor: 9 August 2026 **Evidence window:** 9-20 August 2026 **Dating basis:** Restaurant/human-world work starts in the 9 August commit and continues through major world updates on 18-20 August.

Note: Later project phases overlap because experiments, debugging, writing and analysis happened in parallel. These dates are tracking anchors, not claims that work stopped at phase boundaries.

[AI assistance note: The Gazebo restaurant world was built by Saarunathan with extensive ChatGPT ("Sara") assistance. ChatGPT helped plan the environment, place tables/props and conversational groups, integrate LIRS-HMLG human actors, tune actor waypoints/speeds, edit Gazebo world XML, create collision/ground-truth/map support and debug the world iteratively. Saarunathan ran the simulator and made the final design decisions.]

The story

Nothing in the proposal anticipated this phase. It was never in the plan. And it consumed weeks.

To compare social navigation properly you need a repeatable world: multiple groups, real obstacles, distractors, and - critically - **ground truth you authored yourself**, so the experiment can be scored independently of whatever perception happens to report.

restaurant_testing.world superseded the earlier restaurant_humans.world used in development. A **20 x 15 m room** (walls at $x = \pm 9.9$ m, $y = \pm 7.4$ m) containing five round dining tables at (-4,-1), (-1,4), (2,-2), (6,1), (5,-4); a buffet at (-5.5, -6.0); five plants; a stage at (7.45, 5.20); and kitchen partitions on the west side.

Fifteen human actors arranged into six targets:

Target	People	Centre	O-space radius
Group A	4	(-3.50, -2.50)	0.71 m
Group B	3	(4.67, 2.67)	0.65 m
Group C	5	(5.60, -1.80)	0.87 m
Solo 1	1	(-6.0, -5.0)	-
Solo 2	1	(-6.0, -7.0)	-
Walker	1 (moving)	(4.0, 2.0)	-

Only Groups A, B and C count as valid approach targets (`min_group_size = 2`). A lone individual has no F-formation and no O-space, so approaching one cannot demonstrate the behaviour under study. The solos and the walker exist purely as obstacles and distractors.

Why hand-built rather than procedural? So that every group's true position, membership and O-space radius could be authored precisely and exported as `.groundtruth.json`. That ground truth is what makes the social metrics computable at all - the recorded PLUS-HRI video never carried metric position data to evaluate against. A parametric generator (`generate_social_world.py`) exists for producing variants (a 3-group/9-person unseen layout, and an "adjacent" layout to stress-test whether clustering correctly separates two nearby groups), but the hand-built world is the one used for every reported trial.

The occupancy map Nav2 plans against is generated directly from the world geometry (`world_to_map.py`) rather than by running SLAM - producing an exact, drift-free map independent of a successful mapping run, at the cost of needing regeneration whenever the world changes.

The papers

Reference	Why it belongs here
Hall (1966) and Kendon (1990)	Stop being literature-review concepts and become authored geometry - the O-space radii above are Kendon's construct, measured in metres
Helbing & Molnár (1995) - Social force model	The repulsive-force formalisation of personal space
Lu, Hershberger & Smart (2014) - Layered costmaps	How social constraints can be expressed to a navigation stack
Macenski et al. (2020) - The Marathon 2 (Nav2)	The navigation system being built on

Pagès, Marchionni & Ferro (2016) - TIAGo	The platform
LIRS (2022) - LIRS-HMLG human model library	The source of the human meshes and animations

The tools

Gazebo Classic 11 · Nav2 · PAL TIAGo simulation packages · LIRS-HMLG · tiago_social_worlds · generate_social_world.py, world_to_map.py, extract_world_groundtruth.py, add_person_collisions.py, add_gazebo_ros_plugins.py

What went wrong

Local Windows development stopped being viable. The full ROS 2 + Gazebo + Nav2 + YOLO workload was too slow on a personal machine. This is the phase where serious simulation work moved to the university lab PCs and a standardised Ubuntu 22.04 / ROS 2 Humble VS Code dev container.

Actors had no collision bodies. The human models rendered visually but the LiDAR passed straight through them - the robot could not perceive people as physical obstacles at all.

Furniture was missing from the occupancy map. world_to_map.py originally rasterised only <model> elements using box geometry. The five dining tables are <include> blocks referencing a mesh. **All five were entirely absent from the map**, and Nav2 happily planned paths straight through them.

How it was tackled

- add_person_collisions.py gives every actor a collision body, so the laser sees people.
- <include> elements are now rasterised as 1.2 m squares, taking the map from **9 obstacles to 20**.
- Development standardised on the lab dev container, with docs/LAB_PC_SETUP.md carrying separate setup guidance for the second machine.

What to film / show

A-roll: use a top-down sketch first, then reveal the real Gazebo world. Emphasise that this entire test environment became a major unplanned subproject.

Exact world/assets: src/tiago_social_worlds/worlds/restaurant_testing.world, restaurant_testing.groundtruth.json, restaurant_final.world, restaurant_final.groundtruth.json, plus src/tiago_social_worlds/maps/ and RViz config.

Exact build scripts: scripts/generate_social_world.py, scripts/world_to_map.py, scripts/extract_world_groundtruth.py, scripts/add_person_collisions.py, scripts/add_gazebo_ros_plugins.py, scripts/prepare_lirs_selected_models.py.

Source pages/images: Dissertation pp.43-45 (ROS architecture, Gazebo environment and top-down experimental layout). Show the group table with centres/radii. If possible record a fresh overhead camera pass through restaurant_testing.world.

ChatGPT credit: this is the phase where ChatGPT (Sara) should be visibly credited for extensive world-design/build assistance. If you show a before/after world montage, add a small caption: "Environment built iteratively with ChatGPT-assisted Gazebo/XML guidance; final layout and tests by Saarunathan."

The handoff

"I have models that predict a pose and a world to execute it in. Now every subsystem has to run at the same time."

PHASE 8 - CLOSED-LOOP INTEGRATION

"Everything Works Alone. Together, Everything Breaks."

Date / work window

Documentary anchor: 19 August 2026 **Evidence window:** 19-23 August 2026 **Dating basis:** Closed-loop mission/policy arbitration and reliability fixes dominate commits from 19-23 August.

Note: Later project phases overlap because experiments, debugging, writing and analysis happened in parallel. These dates are tracking anchors, not claims that work stopped at phase boundaries.

The story

The runtime chain:

```
RGB + depth camera
-> person detection (YOLOv8n / LocateAnything-3B)
-> depth back-projection to metric 3D
-> TF: camera_optical -> base_link -> odom -> map
-> clustering into conversational groups
-> policy (rule / Random Forest / MLP)
-> Nav2 goal
-> trajectory + social metrics
```

Six ROS 2 nodes across three packages:

```
src/tiago_group_approach/tiago_group_approach/
group_perception_node.py      detection, depth back-projection, clustering
group_approach_baseline_node.py rule-based policy (geometric)
bc_policy_node.py             learned policy (Random Forest or MLP)
mission_node.py               scripted patrol tour and reporting
gt_localisation_node.py       ground-truth map->odom at 30 Hz
metrics_recorder_node.py      trajectory sampling and social scoring
```

group_perception_node is the moment offline data analysis becomes a robot system. It closes the pixel-to-metres gap that constrained every earlier stage: it back-projects each detection using the pinhole relation $x=(u-cx)d/fx$, $y=(v-cy)d/fy$, $z=d$, transforms into the map frame, and clusters **in genuine world coordinates** with a 1.5 m single-linkage threshold - publishing a real-metres /group_centroid that both policy families consume identically.

The rule baseline is not a straw man. It took six rounds of empirically-driven correction, each one a measured finding:

- Standoff 1.2 m** - deliberately set at Hall's boundary between personal and social space.
- Goal throttling** - publishing a fresh goal every perception frame (~2 Hz) made Nav2 continuously pre-empt and restart planning, so the robot barely moved. A new goal now issues only if the target moved > 0.40 m or the previous goal finished.
- Per-person clearance, not centroid standoff** - a live run measured an intrusion at **0.43 m from the nearest person**, inside Hall's intimate distance, *while satisfying* the centroid standoff, because the centroid shifts when only part of a group is visible. The pose is now checked against every detected person and pushed back until clear of all of them, targeting 0.7 m to the nearest person.
- Body radii, not centre-to-centre** - treating clearance as point-to-point rather than footprint-to-body (TIAGo ~0.30 m radius, a person ~0.25 m) left only **0.15 m of real gap** at a nominal 0.7 m setting. The robot wedged itself into a group, came within **0.062 m** of a person, and stayed stationary for **84% of a ten-minute run**. Both radii are now added explicitly.
- Gap-based approach selection** - evaluate candidate angles around the group and pick one with at least a 60° clear arc, rather than assuming the direct robot-to-centroid line is unobstructed.
- Stuck detection and an "unwedge" reflex** - if the robot hasn't moved 0.10 m in 25 seconds (after a 30 s grace), reverse at -0.15 m/s for 3 seconds and retry.

The papers

Reference	Why it belongs here
Satake et al. - how a robot should approach	Approach <i>direction</i> has social meaning, not just the endpoint
Truong & Ngo - unified approach-pose prediction and socially aware navigation	The closest published framing to this project's exact aim
Repiso et al. (2020) - adapting relative position during interaction	The honest limitation: this project predicts a stopping pose , not a full dynamic interaction policy
Macenski et al. (2020), Lu et al. (2014)	Nav2 and costmap behaviour under real constraints
Hall (1966), Helbing & Molnár (1995)	Now operational - 1.2 m and 0.7 m are literature values, not tuning knobs

The tools

ROS 2 Humble · TF2 · Nav2 (global planner, DWB local controller, costmap_2d, behaviour tree) · Gazebo Classic 11 · Ultralytics YOLOv8n · scikit-learn / joblib · run_everything.sh, run_pipeline.sh, run_trials.sh

What went wrong

This is the phase with the most failures, and they should be shown as a montage.

World and simulator:

- **Stale world file.** Gazebo loads worlds from a copy held by `pal_gazebo_worlds`. Edits to the project's own world file had no effect at all until that copy was refreshed.
- **Missing ground truth.** `restaurant_testing.groundtruth.json` did not exist; the pipeline refused to start.
- **`map\_server` never configured.** It is a lifecycle node whose `configure` step loads the map, and PAL's launch files set no `yaml_filename` for a custom world - so `configure` failed *silently* and the node stayed unconfigured. No `/map`, no map frame, no planning possible.
- **CameraInfo QoS mismatch.** `camera_info` was subscribed with the default **RELIABLE** profile while Gazebo publishes sensor data **BEST EFFORT**. ROS 2 QoS compatibility is one-way: a reliable subscriber silently receives nothing from a best-effort publisher. The topic listed correctly, `count_publishers` returned 1, and no message ever arrived. This was the cause of every `Waiting for CameraInfo...` hang.

Goal arbitration - the one that stopped the experiment dead:

- Both `mission_node` and the active policy sent `NavigateToPose` goals to the same action server, where a new goal pre-empted the old one. Logs showed approach goals accepted and reported "finished" **six milliseconds** later. **The robot never executed a single approach.**
- **No goal throttling in the learned policies.** The fix from rule-policy round 2 had never been ported to `bc_policy_node`, so both learned policies issued a fresh goal every perception frame, each pre-empting the last.
- **Approach never released the mission.** Completion was published the instant Nav2 reported arrival, so the mission resumed immediately and the robot rolled straight past.
- **No memory of attempted groups.** The policy retried the same unreachable group indefinitely - one run lasted **30 minutes, drove 208 m, and was 65% stationary.**
- **Coverage dominated by policy convergence.** The mission yielded to a single approach for up to 45 s with no overall limit. Measured coverage: **34 map cells for the rule policy versus 14-15 for the learned policies**, which never crossed the centre of the room.

Diagnostics:

- **Detection overlay only drawn on processed frames.** Perception runs at 2 Hz, the camera at ~15 Hz. Thirteen of every fifteen frames published no overlay, so the RViz panel appeared frozen - read as "detection is broken" when it was working correctly at its intended rate.

How it was tackled

- Bring-up now **syncs the world file on every run** and reports when the installed copy was stale.
- Ground truth is generated from the world file and regenerated after every world change.
- `yaml_filename` is set explicitly before configuring `map_server`; the project starts its own when PAL navigation is disabled.
- `camera_info` subscribed with the **sensor QoS profile**.
- **Explicit arbitration:** the policy publishes `/approach/start` before driving; the mission cancels its own goal and stands down until `/approach/complete` or a timeout. Arrival is verified **by position** (within 0.75 m of the intended pose) rather than trusting Nav2's status flag, which reports "finished" for aborted goals too. A dwell was added so the robot actually stays.
- Same 2 s / 0.4 m throttle applied to `bc_policy_node`.
- Group positions remembered in 1.5 m cells, retired after three failed attempts or one success.
- A **total approach budget per run**, so the patrol completes regardless of which policy is driving.
- The overlay redraws its cached boxes on every skipped frame and publishes a status banner in every state.

Attempted and explicitly not adopted: TIAGo's base laser scans at 0.2 m and cannot see a tabletop at 0.75 m. Adding the depth point cloud as a second Nav2 observation source via `pointcloud_to_laserscan` would in principle fix that. It destabilised the navigation stack - in one configuration the robot did not move at all - and **was not used for any reported experiment**. It is retained behind a `CAMERA_OBSTACLES=1` flag and named as future work rather than silently dropped.

What to film / show

A-roll: "Every subsystem works in isolation. Now I have to make them agree on one robot, one map and one goal."

Exact ROS nodes: src/tiago_group_approach/tiago_group_approach/group_perception_node.py, group_approach_baseline_node.py, bc_policy_node.py, mission_node.py, gt_localisation_node.py, metrics_recorder_node.py, plus explore_node.py as the earlier optional patrol.

Exact run scripts: scripts/run_everything.sh, run_pipeline.sh, run_trials.sh, preflight_check.sh, check_sim_setup.sh.

Source image: Dissertation Figure 3.7 on p.43 - use this as the main architecture graphic. Animate tokens moving perception -> policy -> Nav2 -> metrics.

Bug montage: stale world, missing map, CameraInfo QoS, mission/policy goal pre-emption, repeated learned goals, no dwell, repeated unreachable groups. Use real terminal/log snippets where available.

The handoff

"The closed loop runs. But there is still an unkept promise sitting in my proposal: LocateAnything-3B."

PHASE 9 - THE PERCEPTION PLAN CHANGES

"The Model I Proposed Isn't Necessarily the Model I Should Deploy"

Date / work window

Documentary anchor: 13 August 2026 **Evidence window:** 13-24 August 2026 **Dating basis:** LocateAnything setup appears 13 August; model-card decoding fix lands 24 August.

Note: Later project phases overlap because experiments, debugging, writing and analysis happened in parallel. These dates are tracking anchors, not claims that work stopped at phase boundaries.

[AI assistance note: The uploaded development record attributes substantial late-stage debugging/analysis assistance from roughly this point onward to Claude. ChatGPT had already assisted the earlier research/planning/ROS stages and the Gazebo world build. Saarunathan performed the experimental runs and retained final decision authority.]

The story

The proposal's **Objective 2 committed to NVIDIA LocateAnything-3B** for people localisation. A vision-language grounding model, ~3 billion parameters, that finds objects from a text query.

By implementation time, one thing had become obvious: perception has to run **continuously inside a navigation loop**. A robot that sees the world once every several seconds is not navigating socially; it is navigating blind between snapshots.

So the objective changed - and a changed objective needs evidence, not a shrug.

YOLOv8n (~3.2 M parameters) reached **99.7% recall** on the annotated sessions, comfortably above the 80% target. A direct 30-frame comparison gave **29/30 for YOLO** and **30/30 for LocateAnything**, with substantially overlapping confidence intervals - LocateAnything is marginally more accurate.

The difference is speed, and it is not marginal.

The papers

Reference	Why it belongs here
Redmon et al. - YOLO	The single-stage real-time detection foundation
Jocher, Chaurasia & Qiu (2023) - Ultralytics YOLOv8	The actual implementation used
Yaseen - YOLOv8 architecture analysis	Supporting characterisation
Ren et al. - Faster R-CNN	Reviewed alternative: two-stage, more accurate, too slow
Carion et al. (2020) - DETR	Reviewed alternative: transformer detection, end-to-end
Wang et al. (2026) - LocateAnything	The proposed model, and the model card that eventually solved the bug

Reviewing Faster R-CNN and DETR rather than naming one model without justification is a **direct response to the Report 1 supervisor feedback (AP4)**. This is the *second* feedback round and should not be confused with the proposal feedback in Phase 2.

The tools

ultralytics/ultralytics · NVIDIA LocateAnything-3B via HuggingFace · PyTorch + CUDA · a **separate `la3b\_env` virtual environment**, because LocateAnything's dependencies conflict irreconcilably with the main NumPy/scikit-learn environment · locateanything_service.py (a local HTTP inference service)

What went wrong

341 bounding boxes on a single frame. The model was emitting hundreds of near-identical boxes per image, taking around ten seconds per frame and producing unusable output.

One detection per entire trial. The original integration ran LocateAnything in "one-shot" mode - look once, then never again. Under that mode a whole trial's behaviour derived from a single glance at the room. That is not a policy comparison; it is a comparison of three policies acting on stale information.

Boxes clipped at the image edge. A person half-out of frame produces a bounding box whose centroid is meaningless, which then poisons the group centroid.

How it was tackled

The 341-box bug was solved by actually reading the model card. Two mistakes, both mine to find:

1. The wrong chat-template method was being called. The model ships `py_apply_chat_template`; the code used the generic one.
2. **Greedy decoding.** An autoregressive decoder run greedily can enter a repetition loop and emit the same box forever. The card specifies sampling.

The fix: `py_apply_chat_template` (via `getattr` with a fallback), `do_sample=True`, `temperature=0.7`, `top_p=0.9` with `torch.manual_seed(0)` for reproducibility, and `max_new_tokens=1024` to bound the output regardless. Plus a `_sanitise()` guard - `MAX_BOXES=20`, `MIN_AREA_FRAC=0.0005`, `MAX_AREA_FRAC=0.60`, `DEDUP_TOL_PX=8.0` - so a degenerate generation can never again reach the policy.

Inference dropped from roughly ten seconds per frame to about 0.4 seconds.

One-shot was replaced with a threaded periodic mode. `group_perception_node` gained a third mode alongside `continuous` and `oneshot`: `dispatch` inference to a worker thread on a timer, `debounce` so a slow inference cannot queue up behind itself, and keep publishing while the worker runs. The TF buffer cache was extended to 30 seconds so a detection produced 0.4 s after its frame can still be transformed correctly.

Edge-clipped boxes are rejected within `edge_margin_px` (default 8) of the image border.

⚠ The timing number - handle carefully on camera

There are **three** `LocateAnything` inference figures in the project record, and they describe three different configurations:

Source	Mean per frame
Stored benchmark artifact (<code>comparison_manifest.csv</code> , 9 Aug)	25.63 s (median 24.38, range 21.6-42.2)
Submitted dissertation benchmark	8.40 s (median 9.64)
After the model-card fix (24 Aug)	~0.4 s

The dissertation's "approximately 1,700x slower than YOLO" rests on the 8.40 s figure. After the fix it is closer to 80x.

Do not put a multiplier in the spoken cut until the exact service build used for each final trial has been reconstructed. The safe line:

"My first implementation was extremely slow. I found I was calling the model incorrectly and made it dramatically faster - but YOLO was still the practical real-time detector for a closed navigation loop."

What to film / show

A-roll: explain the difference between "best model on paper" and "model that can keep a moving robot informed".

Exact files: `scripts/locateanything_service.py`, `scripts/test_locateanything_detector.py`, `scripts/run_locateanything_comparison.py`, `scripts/export_detector_comparison_frames.py`, `requirements-locateanything.txt`, `la3b_env/`, `yolov8n.pt`.

Source pages/images: Dissertation pp.33-35 detector alternatives/validation. Show the 341-box failure if you have a capture. Then show the model-card call and periodic threaded mode.

Important spoken-rule: do not use a `LocateAnything` speed multiplier unless you explicitly label the configuration (25.63 s early artifact, 8.40 s submitted benchmark, ~0.4 s after the 24 Aug fix).

The handoff

"The detector question is settled. Then I ran the first proper experiments and found something much worse than a slow model: my own measurements were lying to me."

PHASE 10 - THE MEASUREMENT CRISIS

"The Robot Wasn't Failing. My Experiment Was."

Date / work window

Documentary anchor: 21 August 2026 **Evidence window:** 20-23 August 2026 **Dating basis:** Measurement/localisation/collision/task-scoring fault fixes cluster on 20-23 August.

Note: Later project phases overlap because experiments, debugging, writing and analysis happened in parallel. These dates are tracking anchors, not claims that work stopped at phase boundaries.

The story

This is the phase where the project stops being software engineering and becomes science.

Fault 1 - Localisation. The map->odom transform was published **once, statically, at startup**. It is correct for an instant and diverges progressively afterward, especially after wheel slip - which collisions reliably cause.

This was not cosmetic. It **corrupted the map-frame position of every detection made under that scheme**. One detection at (4.4, 2.8) was initially misdiagnosed as a YOLO false positive. It was a *correct* detection of a real person at (-3.0, 0.0), rendered nonsense by a stale transform.

Replaced with `gt_localisation_node`, recomputing and republishing $T_{map_odom} = T_{map_base} \cdot T_{odom_base}^{-1}$ continuously at **30 Hz** from simulator state. Detections then landed **within 10 cm of ground truth**.

This is declared in the dissertation as a deliberate methodological choice: localisation comes from simulator ground truth **specifically so navigation error cannot confound the comparison between policies**. Laser, costmaps and planning all still operate normally through Nav2.

Fault 2 - The collision metric. Every trial in the first full experiment reported a collision. Nineteen runs, `min_obstacle_range_m` between **0.200 and 0.267 m**, twelve of them at exactly 0.200 m.

That is not nineteen collisions. That is a constant. The base laser was returning **the robot's own chassis**.

With a 0.30 m collision threshold this flagged a collision in every trial - and because task success is gated on collision-free operation, **success was measured as 0% for all three policies for reasons entirely unrelated to their behaviour**.

Fault 3 - Task success scored against a phantom. `task_success` was evaluated against `goal_centroid`, a variable overwritten by every incoming `/group_centroid` message. It therefore held whatever perception saw *last* - frequently a false positive near a wall.

One MLP trial came within 0.51 m of a real person and registered a genuine O-space intrusion, yet scored as never having held a valid pose - because it had approached a real group correctly and was then judged against a phantom.

Fault 4 - A protocol flaw that inverted a conclusion. Before `min_group_size` was fixed at 2, an earlier batch of 61 trials scored approaches to targets of *any* size, including lone individuals:

Policy	Approached a real (2+) group	Approached only lone individuals
Rule-based	12/12 (100%)	0
BC - Random Forest	21/26 (81%)	4
BC - MLP	4/23 (17%)	11

The MLP was not being socially cautious. It was being handed lone individuals by perception, and its apparently excellent O-space score was largely an artefact of rarely approaching a real group at all.

The papers

Reference	Why it belongs here
Hoffman & Zhao (2020) - <i>A primer for conducting experiments in HRI</i>	The paper that matches this phase exactly. Experimental validity as a first-class concern
Fisher (1922)	Fisher's exact test - the correct statistic for 10-trial binary comparisons, where χ^2 is not valid
Mavrogiannis et al. (2023) - Core challenges of social robot navigation	Why proxy metrics for social behaviour are hard to get right
Francis et al. (2025) - Guidelines for evaluating social robot navigation	Evaluation methodology as its own research problem

The tools

metrics_recorder_node.py · rescore_sim_results.py · gt_localisation_node.py · SciPy (Fisher exact) · full 10 Hz trajectory logging

What else went wrong

The full fault record runs to **22 material faults**. The remainder:

- **Gap selection chose the far side of the group.** Candidates were sorted by width alone, so the robot repeatedly picked an opening **175-179° around the group** - walking around, and effectively through, the people to reach it. Logged verbatim: *"177 deg gap ... 179 deg off the robot's current side."* On a five-person test with the robot approaching from the west, the old rule sent it to (7.31, -1.41), **149° around the group**; the new rule sends it to (3.64, -1.19), **0° off**.
- **`num\_people` frozen at inference.** Hard-coded to the constant 3.0, with a comment promising it would be "refined below if perception reports it" - a refinement that never happened. One of seven trained features was a constant at deployment while training data spanned 1 to 6. (Feature importance 0.055 - real but modest, not catastrophic.)
- **Predictions rejected rather than clamped.** An implausible prediction caused the policy to send **no goal at all**, so the robot stood still - easily misdiagnosed as "the model isn't working."
- **Recovery that made things worse.** The stall reflex reversed blindly; TIAGo's laser covers only the forward arc, so the robot repeatedly backed into furniture it could not see. A second, related fault disabled the reflex precisely when the robot had been stuck longest, because it keyed off "a goal was recently *sent*" rather than "a goal is currently *in flight*."
- **Arm tuck reported success it had not verified.** `play_motion2` home was used to stow the arm before each trial. home **extends** the arm rather than tucking it - and the check treated "the command returned exit code 0" as proof the arm had moved.
- **Trials starting from the previous trial's end pose.** Because the simulation stays running across trials for efficiency, each trial began wherever the last one ended. One trial's very first trajectory sample was already at (-2.46, -2.48) - standing among people before the policy had acted at all.

How it was tackled

- **Collision:** calibrate the self-hit radius during a startup grace period and discard returns below it. Collision-free rate rose to 100% under normal operation.
- **Task success:** scored against **ground-truth groups**, with distance and heading required to hold **in the same trajectory sample**. Re-scoring the existing trials raised MLP from 20% to 60% and Random Forest from 60% to 90%.
- **Gap selection:** any gap of at least 60° is adequate; choose the **nearest** adequate gap, not the widest.
- **`num\_people`:** taken from the live detection count.
- **Predictions:** over-long predictions scaled back onto a plausible range rather than discarded.
- **Recovery:** checks rear beams and rotates in place if the space behind is not confirmed clear; keyed to goals in flight.
- **Arm:** correct `tuck_arm` motion, controller commanded directly, and `/joint_states` read back to verify the arm actually reached the tucked configuration.
- **Start pose:** the robot is explicitly teleported to (0, 0) at the start of every trial.
- **Protocol:** valid approach targets restricted to groups of two or more for every reported result.

The single most valuable design decision in the whole project shows up here. Because every trial's **full 10 Hz trajectory was retained**, all of these metric corrections could be applied by re-scoring recorded runs (`rescore_sim_results.py`) rather than re-running the simulation. Weeks of compute saved by a logging decision made months earlier.

What to film / show

A-roll: this should feel like a detective episode. Start with "every policy failed" on screen, then reveal each measurement fault.

Exact files: `src/tiago_group_approach/tiago_group_approach/gt_localisation_node.py`, `metrics_recorder_node.py`, `scripts/rescore_sim_results.py`, `scripts/run_localisation.sh`, `scripts/start_localisation.sh`.

Graphics: (1) stale map->odom transform drifting; (2) LiDAR ray hitting TIAGo's own chassis; (3) `goal_centroid` changing from a real group to a phantom. Use red stamps "MEASUREMENT ERROR" rather than "ROBOT FAILURE".

Source material: `docs/PROJECT_RECORD_POST_REPORT1.md`, dissertation live-metric sections pp.45-53. Keep the 22-fault list as a rapid montage rather than narrating every item in full.

The handoff

"The measuring instrument is finally trustworthy. But the models still hit a ceiling I can't explain. Maybe the problem was never the model."

PHASE 11 - V2

"I Tried to Improve the Model and Found the Problem Was the Data"

Date / work window

Documentary anchor: 24 August 2026 **Evidence window:** 24 August 2026 **Dating basis:** Periodic LocateAnything mode, V2 study and 360-model search committed 24 August.

Note: Later project phases overlap because experiments, debugging, writing and analysis happened in parallel. These dates are tracking anchors, not claims that work stopped at phase boundaries.

The story

Before building anything, seven candidate improvements were listed and individually assessed:

#	Proposed improvement	Verdict
1	Collect more independent demonstrations	Not possible. PLUS-HRI is a fixed 24-session corpus; new collection needs fresh ethics and participants
2	Train on the whole approach, not the final metres	Implemented - this is V2
3	Add metric social/geometric features	Partly. Metric distances are <i>not derivable</i> - people exist only as pixel boxes from an uncalibrated camera. Angular equivalents added instead
4	Reduce train-to-live feature mismatch	Implemented - every new feature is an angle or a ratio, never a raw pixel value
5	Sample by event, not by row	Implemented - event_id, inverse-frequency weighting, 10 Hz decimation
6	Keep the MLP small and regularised	Already true - the 32/16, $\alpha = 0.1$ network already beat the 128/64 spec
7	Try a stronger tabular regressor	Implemented - HistGradientBoostingRegressor

Then the diagnosis. Reconstructing all 462 V1 events:

Property of a V1 "approach event"	Value
Median distance from event start to stop pose	0.17 m
Median event duration	2.5 s
Events starting more than 1.0 m from the stop	53 / 462 (11%)
Events starting more than 2.0 m from the stop	13 / 462 (3%)
Rows within 0.5 m of the stop	72%
Rows beyond 2.0 m from the stop	3.6%

The typical "demonstration" was a 2.5-second shuffle ending 17 cm from where it began. Not an approach. A final adjustment.

The root cause sits in `find_approach_events()`: it pairs each *moving segment* with the stop that follows, where "moving" is any 1.0 s bin above a 0.05 speed threshold. A human walking toward a group naturally pauses, turns and hesitates - so the walk fragments into many short segments, and **only the final fragment before the stop was ever labelled.**

That is the single best explanation for the live behaviour: sensible predictions within roughly a metre of a group, unreliable ones further out, because the models had barely seen "further out."

V2. `build_approach_pose_dataset_v2.py` anchors on each sustained stop near a group and walks *backwards*, absorbing brief pauses, until it has accumulated at least 1.0 m of travel (or hits a stop longer than 3 s, or exhausts a 20 s lookback).

	V1	V2
Events	462	182

Rows	70,555	128,506
Median event start distance	0.17 m	1.37 m
Median event travel	-	1.98 m
Median distance-to-go	0.183 m	0.784 m
Rows within 0.5 m of stop	72%	35.9%

Six new angular/ratio features were added (group_span_rad, nearest_person_span_rad, gap_bearing_rad, gap_width_rad, person_spacing_rad, people_visible), and the original seven retained unchanged so V1-trained models could be scored on V2's more honest rows.

And it got worse

All models compared on the **same** held-out sessions (5, 9, 59):

Panel	Models scored	Best position (m)	Best orientation (°)	Best "both"
A - V1 models, V1 test rows	reproduces the Phase 6 table exactly to 4 d.p.	0.305 (rule)	25.78 (RF)	43.3%
B - V1 models, V2 (genuine-approach) test rows	V1 RF, V1 MLP, naive, rule	0.652 (V1 MLP)	34.91 (V1 RF)	23.5%
C - V2 models, V1 features only	RF, GB, MLP, all V2-trained	0.722 (RF)	34.93	15.5%
D - V2 models, V2 features	as C + six new features	0.722 (RF)	36.55	15.5%

Re-segmentation alone, and re-segmentation plus new geometry, both perform worse than simply keeping the V1 model and testing it on genuine approaches. The attempted improvement improved nothing.

The finding hiding inside the failure

Moving from V1's terminal-adjustment test rows (Panel A) to V2's genuine-approach rows (Panel B) **roughly doubles every policy's position error** - naive 0.410 -> 0.724 m, tuned Random Forest 0.365 -> 0.656 m - and the joint-threshold pass rate falls from 43.3% to 23.5%.

The published offline evaluation was, in effect, being scored on the easiest part of the task. Found and reported by the author, not by an examiner.

And more strikingly, **the ranking between rule and learned reverses:**

On terminal adjustments (V1 rows)	On genuine approaches (V2 rows)
Rule wins on position: 0.305 m vs RF's 0.365 m	Rule becomes the worst policy tested : 0.765 m - beaten even by predicting the training mean (0.724 m)
	Random Forest becomes the best : 0.656 m, 23.5% within both thresholds

A fixed geometric rule - face the group, stop a fixed distance short - is adequate for the last half-metre and degrades as soon as there is a real approach to plan.

Three follow-up studies

Does more training data recover the loss? Partially. Sweeping the minimum-travel threshold, test set and learner held fixed:

Configuration	Train events	Position (m)	Orientation (°)	Within both
min_travel = 1.0 m (V2 default)	120	0.722	34.93	15.5%
min_travel = 0.5 m	178	0.657	34.69	19.6%
min_travel = 0.25 m	216	0.648	35.22	21.9%
0.25 m + mirror augmentation	432	0.652	34.01	22.1%

0.25 m + mirror + validation folded in	578	0.642	33.56	24.2%
reference: shipped V1 Random Forest	-	0.656	34.91	23.5%

Position error falls **monotonically** with event count - the re-segmentation was sound, the corpus was simply too small to support it. Reported with three honest qualifications: the gain over the shipped model is marginal (2.1%); **mirror augmentation is nearly inert** (doubling 216 -> 432 events moved error 0.004 m in the *wrong* direction, showing the shortage is behavioural variety, not row count); and the 578-event figure is mildly optimistic because those hyperparameters were originally selected using a search that saw the validation split.

Extrapolated: 4.8x more events bought an 11% error reduction. Reaching the 0.4 m threshold would need substantially more independent demonstration data than 24 sessions contain - turning "collect more data" from a generic suggestion into a **quantified projection**.

Does a higher frame rate help? No - monotonically worse.

Training frame rate	Train rows	Position (m)	Orientation (°)	Within both
2 Hz	5,883	0.640	34.25	22.7%
5 Hz	14,016	0.643	34.62	22.0%
10 Hz (default)	26,194	0.648	35.22	21.9%
20 Hz	43,967	0.653	35.22	21.3%
all rows (~33 Hz)	117,857	0.667	35.90	20.1%

Twenty times the rows costs 4.3% in position error. At 33 Hz the robot moves about a centimetre between samples and the group hasn't moved at all - each extra row is a near-duplicate that dilutes rather than strengthens the bootstrap statistics.

Read together, this is the key methodological result: what matters is the number of independent demonstrated **events** (120 -> 578 improved error 11%), not the number of **rows** (5,883 -> 117,857 made it 4% *worse*). Which retrospectively corrects how the dataset should be described - **"70,555 rows" overstated the evidence**. The true sample size is 462 events, 182 of them genuine.

Does the model underfit, or does the label set a floor? The label.

	Position error
Predict the training-set mean	0.746 m
Random Forest on its own training rows	0.597 m
Random Forest on held-out test rows	0.648 m

An unconstrained forest reaches 0.210 m on training data - the model plainly *can* fit it - but every capacity increase made **test** error worse (0.648 -> 0.739 m unconstrained).

Then the measurement that explains everything: **for each training row, its ten nearest neighbours in standardised feature space disagree on their labelled stop pose by a mean of 0.505 m**. Rows the model cannot distinguish specify stop poses half a metre apart.

That is a **label-ambiguity floor**, and it is a direct consequence of the target definition: `target_dx`, `target_dy` is displacement from the robot's *current* pose, so two moments with identical group bearing, apparent size and LiDAR range can sit at very different points along an approach - one two metres out, one thirty centimetres out - carrying completely different labels.

A group-frame reformulation was tried, and failed instructively. If the target is ill-posed because it is measured from the robot, measure it from the group:

Target formulation	Train pos.	Test pos.	Test "both"
Robot-frame displacement (shipped)	0.597 m	0.648 m	21.9%
Group-frame standoff + bearing	0.651 m	0.747 m	15.2%

Worse. Reconstructing a group-frame goal requires knowing how far away the group actually is, and the only estimate available is `lidar_min_range` - the same documented proxy already used as a feature. Its error now enters twice: once building the label, once converting a prediction back. **The root cause is a property of the corpus, not the method**, and no choice of target formulation or architecture can repair it.

The 360-model search

`finetune_on_lab_pc.py`: **120 randomly sampled configurations for each of three families - 360 models** - selected on the validation sessions, scored once on the held-out test sessions, with ranges deliberately bracketing the shipped values in both directions so the search could pick a *smaller* model too.

Model family	Position (m)	Orientation (°)	Within both
Random Forest	0.636	34.05	24.8%
Gradient Boosting	0.631	33.66	24.9%
MLP	0.627	33.55	26.1%
<i>shipped V1 Random Forest</i>	0.656	34.91	23.5%
<i>label-ambiguity floor</i>	0.505	-	-

Three observations matter more than the 0.029 m improvement:

- All three families finish within 0.009 m of each other.** A random forest, a boosted ensemble and a neural network share no architectural assumptions. Their convergence on the same number is independent confirmation that the limitation is the data.
- Only 19% of the available headroom was captured.** The gap to the 0.505 m floor is 0.151 m; exhaustive search recovered 0.029 m of it.
- Every family independently chose heavy regularisation** - winning RF `max_depth=4`, `min_samples_leaf=40`; Gradient Boosting `max_depth=2`; MLP `α=10.0`. Three independent searches all concluding smaller is better.

Notably the **MLP - the weakest model in the original evaluation - is the best of the three once properly tuned**. Its earlier underperformance was a tuning artefact, not an architectural limitation. (Its winning configuration did emit a `ConvergenceWarning` at `max_iter=400`, so that figure carries small residual uncertainty. Stated, not hidden.)

And the tuned models were not deployed for the offline claims. The improvement is real but small, and Section 3.6 already establishes that offline position error does not predict live social behaviour. The study's value is **evidential, not operational**: it confirms with a properly powered search that the shipped configuration is within 4.4% of the best findable, and that the residual error belongs to the dataset.

The papers

Reference	Why it belongs here
Ross, Gordon & Bagnell - DAGger	Distribution shift: a cloned policy visits states its demonstrations never covered. Exactly the terminal-adjustment problem
Torabi et al. - imitation from observation	Learning when the demonstrations are incomplete
Codevilla et al. (2019) - <i>Exploring the limitations of behavior cloning</i>	The paper that names this failure mode. Required reading for this phase
Ravichandar et al.	Demonstration coverage and data efficiency
Friedman (2001) - Greedy function approximation	The gradient-boosting foundation
Ke et al. (2017) - LightGBM	The histogram-binning strategy behind <code>HistGradientBoostingRegressor</code>
Breiman (2001)	Random Forest, again

The tools

`build_approach_pose_dataset_v2.py` · `train_evaluate_v2.py` (four-panel comparison, resumable via cached joblib) · `sweep_dataset_v2.py` · `sweep_decimation.py` · `finetune_on_lab_pc.py` · `scikit-learn`

One bug worth showing

In `finetune_on_lab_pc.py`, sampling `max_features` from a mixed list `[1.0, 0.7, 0.5, "sqrt"]` via `rng.choice` **silently coerced every value to ``numpy.str_``** - so a numeric fraction became the string `"0.7"` and `scikit-learn` rejected it. Fixed by indexing instead:

```
max_features=[1.0, 0.7, 0.5, "sqrt"][int(rng.integers(4))],
```

A one-line bug that would have invalidated 120 of the 360 models.

What to film / show

A-roll: put 0.17 m on a whiteboard and ask: "Does this look like an approach?" Then reveal 1.37 m after V2.

Exact files: scripts/build_approach_pose_dataset_v2.py, train_evaluate_v2.py, sweep_dataset_v2.py, sweep_decimation.py, finetune_on_lab_pc.py; docs/V2_RETRAINING_STUDY.md.

Visuals: V1 vs V2 event diagram, 120 -> 578 independent events, 5,883 -> 117,857 rows, 0.505 m label-ambiguity floor, and a 360-model grid collapsing to RF/GB/MLP within 0.009 m.

Actor beat: do not call V2 a success. The emotional payoff is "I fixed the task definition, made the evaluation harder, and the evidence told me the limitation was the demonstrations."

The handoff

"I understand the learning ceiling now. But one formal objective still hasn't been tested at all."

PHASE 12 - CLOSING THE EVIDENCE

"The Objective I Still Hadn't Proven"

Date / work window

Documentary anchor: 25 August 2026 **Evidence window:** 25 August 2026 **Dating basis:** Objective 3 O-space validation committed 25 August.

Note: Later project phases overlap because experiments, debugging, writing and analysis happened in parallel. These dates are tracking anchors, not claims that work stopped at phase boundaries.

The story

Objective 3 asked whether the group/O-space estimate matches human judgement. Validating it required a hand-labelled ground truth **that does not exist natively in the dataset** - so one was built, in four stages, each its own script:

1. ``extract_clean_ospace_frames.py`` selected 30 frames spanning 18 sessions, each containing a genuine group of two or more, avoiding motion-blurred or corrupted frames.
2. ``label_ospace_frames.py`` with a purpose-built browser tool (`label_ospace.html`) presented each frame to a human labeller, who clicked the point they judged to be the true O-space centre.
3. ``merge_ospace_clicks.py`` merged the click coordinates back into `labels.csv`, matching by `frame_file` rather than row order so a session could be paused and resumed without misaligning rows.
4. ``validate_ospace_estimate.py`` compared each manual label against the pipeline's automatic centroid.

The frames deliberately carry no overlay. The earlier `prepare_ospace_validation.py` burned the detector's boxes and a red cross marking the *estimated* centre into the pixels. Useful for eyeballing, unusable as a labelling surface: a human asked to mark the true centre with a red cross already sitting there will anchor to it, and the "validation" would measure agreement with the algorithm rather than accuracy against human judgement.

Two failures in the AI-assisted labelling workflow

The labelling instruction was wrong, and the data proved it. The first run asked labellers to click the group's centre **at floor level** - the natural reading of "where the group is standing." It produced a **161-pixel systematic bias with a standard deviation of 48 px**, and - the giveaway - **all 30 clicks fell on the same side.**

That is not labelling noise. A consistent one-directional offset is a definitional mismatch. The pipeline's estimate is built from detection **bounding-box centres**, which sit at roughly chest height. A floor-level click is geometrically a different point, offset by the perspective foreshortening between a standing person's feet and their torso as seen from an elevated, angled camera.

The instruction was corrected to click at **body height**, and the study re-run. `labels_floor_definition.csv` is retained alongside the corrected `labels.csv` **specifically so the correction remains auditable** rather than silently overwritten.

The labelling tool itself had a bug. The first corrected run recorded **29 of 30 frames at the identical pixel (234.3, 253.5)** - the tool advanced too eagerly, so a single click registered across many frames. Fixed with a `MIN_CLICK_INTERVAL = 0.6` second debounce and a `--restart` flag.

Both problems occurred in the Claude-assisted labelling workflow and are retained here because a documentary about honest failure should include failures introduced by supporting tools as well as failures in the robot system.

The criterion had to change, and that was disclosed

The proposal's original criterion - within **0.3 m** of the manual label for $\geq 70\%$ of frames - **could not be evaluated honestly.** 0.3 m is a metric tolerance and the video carries no depth or calibration from which real metres can be recovered. Reporting a metre figure would have been fabricated.

Re-specified **before use** as: within **0.5 x the mean detected person bounding-box width**, for at least 70% of frames. An adult's shoulder width is roughly 0.45-0.50 m, so 0.5x bounding-box width corresponds to approximately **0.22-0.25 m** of real tolerance - preserving the original order of magnitude while expressing it in a unit this dataset can actually measure, and one that self-corrects for perspective (a person further away gets a proportionally tighter pixel tolerance, which is physically correct).

The result

Quantity	Value
Frames prepared / scored	30 / 30 (none excluded)
Tolerance	0.5 x mean person bounding-box width
Mean error	92.36 px (0.760 person-widths)

Median error	75.11 px
Frames within tolerance	11 / 30 (36.7%)
Target	$\geq 70\%$
Verdict	FAIL

`validate_ospace_estimate.py` carries an explicit instruction in its own source: *"A fail here is still a reportable result - it quantifies how well a centroid-based O-space estimate matches human judgement, which is exactly what Objective 3 asked you to measure. Report it honestly."*

Two mitigating facts, reported alongside rather than instead: the mutual-facing centroid estimate is independently justified by Vascon et al. (2016), not an arbitrary guess; and the live results that actually depend on O-space accuracy - intrusion rate, task success - are computed against the **world's true ground truth**, not this offline pixel estimate, so this failure does not propagate into and invalidate the live experiment.

The writing - which was never a separate final step

This is also where the documentary should finally show the writing, because it was never "finish the robot, then write 93 pages." It was:

read -> design -> build -> discover problem -> read again -> change method -> experiment -> rewrite

The bibliography is the evidence. **The proposal cited 7 references. The final dissertation cites 38.** Each expansion was forced by a problem:

Problem hit	Literature it forced
No orientation data	Vascon, Cristani, Swofford - position-only group detection
Learned models beaten by a mean predictor	Codevilla, Ross/DAGger, Torabi - BC limitations
Needed a stronger regressor	Friedman, Ke - boosting
Detector substitution needed justifying	Redmon, Ren, Carion, Jocher
Nav2 and costmap behaviour	Macenski, Lu, Helbing & Molnár
Metrics that turned out invalid	Hoffman & Zhao, Fisher, Mavrogiannis, Francis

Final document: 93 pages, six chapters, 38 references.

And the documentation pass

Separately from the dissertation, the codebase itself was made readable end to end: every emoji removed, every `#!/usr/bin/env python3` stripped, a plain-language "what this is / why it exists / how to run it" header added to **every** script and node, thin headers expanded, and the README rewritten (~2,100 words) with the restaurant-waiter framing, proxemics and F-formations explained from scratch, a pipeline diagram, the feature table and working figure links. Then a submission package with a `CONTENTS.md` manifest mapping every reported claim to its source file.

What to film / show

A-roll: record the manual labelling process. This is a good quiet late-project scene: clicking frames, checking CSVs, writing.

Exact files: `scripts/extract_clean_ospace_frames.py`, `scripts/label_ospace_frames.py`, `label_ospace.html`, `scripts/merge_ospace_clicks.py`, `scripts/validate_ospace_estimate.py`, `scripts/prepare_ospace_validation.py`.

Source pages/images: Dissertation pp.76-77 O-space result; show 11/30 = 36.7% against $\geq 70\%$ on a simple scoreboard. Also show the 93-page dissertation thumbnails and references growing from 7 in the proposal to 38 in the final dissertation.

Writing montage: `Introduction_Draft.md`, `Literature_Review_Draft.md`, `Methodology_Draft.md`, `Implementation_Draft.md`, `Results_Discussion.md`, `Conclusion_Draft.md`, `References_Draft.md`, `README` and docs files.

The handoff

"Everything is measured, explained and written. One question left: what does the finished experiment actually say?"

PHASE 13 - THE FINAL ANSWER

"Learning Didn't Win. Rules Didn't Win Either."

Date / work window

Documentary anchor: 24-25 August 2026 **Evidence window:** 24-30 August 2026 **Dating basis:** Final trial set and slot-accuracy analyses were completed/committed 24-25 August; final submission/write-up window continued to end of August. Demo archive commit followed 6 September.

Note: Later project phases overlap because experiments, debugging, writing and analysis happened in parallel. These dates are tracking anchors, not claims that work stopped at phase boundaries.

Full beard. Final filming stage.

The protocol

Parameter	Value
Environment	restaurant_testing.world, 20 x 15 m
People	15 actors: 3 groups (4, 3, 5), 2 lone individuals, 1 walker
Valid targets	Groups of ≥ 2 only
Trials	10 per policy per detector - 60 total (3 policies x 2 detectors x 10 repeats), 180 group visits
Patrol route	(-5,5) -> (3,6) -> (8,1) -> (8,-6) -> (-8,-4) -> back to (-5,5)
Localisation	Simulator ground truth at 30 Hz
Trajectory sampling	10 Hz, fully retained
Success criterion	Distance $\in [0.5, 2.0]$ m of a real group centre and heading within 45° , in the same sample
Termination	Route complete, 60 s of no motion (after 90 s grace), or a 30-minute hard timeout
Start pose	Teleported to (0, 0) every trial

A fixed route rather than free exploration, so every trial covers identical ground in identical order - any measured difference reflects the policy, not which part of the room the robot happened to visit.

The binary metrics saturate

Under YOLOv8n, with the tuned models (bc_ft, mlp_ft):

Metric	Rule	BC-RF	BC-MLP
Task success	10/10	10/10	10/10
Collision-free	10/10	10/10	10/10
O-space intrusion	9/10	9/10	9/10
Cut-through	0/10	0/10	0/10
Mean nearest-person distance	0.277 m	0.391 m	0.347 m
Mean path length	67.93 m	57.27 m	56.94 m

Everything passes. Which is exactly the problem. The binary metrics cannot answer *how good* the joining position was, only whether the robot got roughly near a group without hitting anything.

Two things worth narrating anyway: **BC-Random Forest keeps the most distance from people** (0.391 m vs the rule's 0.277 m), and **both learned policies travel about 10 m less** than the rule.

 **Provenance note - say this on camera or caption it.** The dissertation's binary table reports different figures (rule 10/10, BC-RF 10/10, BC-MLP 4/10; MLP clearance 1.01 m; intrusion 10/7/3). Those come from an **earlier experiment using the original `bc` / `mlp` models, around 22 August** - confirmed by matching the distinctive clearance values ($bc = 0.407\text{ m} \sim 0.41$; $mlp = 1.007\text{ m} \sim 1.01$). The table above is

the later `bc\_ft` / `mlp\_ft` run of 24-25 August. Two model generations, two experiments. Label whichever you show.

*The difference is itself a good story: the **untuned MLP was over-cautious** - it stopped a full metre away, so it rarely intruded (3/10) but often failed the task (4/10). Tuning brought it in to 0.347 m, where it succeeds 10/10 and lands 8.5 cm from an ideal slot. Fine-tuning turned a timid model into an accurate one.*

The metric that finally discriminated

Since success saturated, a continuous measure was needed: **how close did the robot come to a socially plausible place to join?**

`approach_accuracy.py` derives, for each ground-truth group:

1. Every member's bearing from the group centre;
2. Angular **gaps** between adjacent members ($\geq 45^\circ$ counts as a real opening);
3. **Ideal joining slots** placed in those gaps at `ospace_radius + 0.6 m` - outside the conversation, in the opening, at conversational distance. Twelve slots per configuration.
4. The robot's closest approach to any slot, **and its final heading**, scored separately.

This is the documentary's best structural payoff: **Kendon's F-formations, encountered in Phase 1 as a paper, become the instrument that measures the final result.**

Two scoring bugs I introduced, and had to fix

Bug 1 - the fly-past. The first version checked heading at the *closest* sample. A robot could sail past a perfect slot at speed, clip the tolerance for a tenth of a second, and score as having reached it. Result: 0/30 reached, with a worst-case error of 0.446 m *inside* a 0.5 m tolerance - nonsense on its face. Fixed by requiring **distance and heading to hold in the same sample**.

Bug 2 - scoring the robot mid-drive. Even then, samples taken while the robot was still moving counted. Fixed with a `stationary_speed <= 0.10 m/s` filter, so only a robot that has actually *stopped* somewhere can be credited with standing there.

The headline result

Under YOLOv8n, recomputed independently from the raw trial files:

Policy	Median slot-position error	Mean orientation error
BC-MLP	0.085 m	79.1°
BC-RF	0.241 m	74.7°
Rule	0.261 m	64.6°

The two columns rank in opposite orders.

And it is not one lucky detector. Under LocateAnything-3B, independently:

Policy	Median slot-position error	Mean orientation error
BC-MLP	0.178 m	79.4°
BC-RF	0.188 m	64.1°
Rule	0.252 m	62.0°

Same position ordering. Same orientation ordering.

The stress test

Under the slower LocateAnything perception, one difference is stark:

Policy	Group cut-through
Rule	6/10 trials
BC-RF	1/10
BC-MLP	0/10

Phrase this conservatively on camera:

"When perception updates became much slower, the geometric rule became much less robust to stale group information and cut through conversational groups in six of ten runs."

Not "LocateAnything proves learning is better." The detector, policy and navigation loop all interact.

The answer

Not "AI beat the rule." Not "the rule beat AI."

The non-expert demonstrations contained genuinely useful information about where the robot should stand - the learned policies place it three times more accurately. But the hand-coded geometric rule remained better at deciding which way to face, because it turns toward the group by construction and cannot be badly wrong, while the learned models must predict the facing and that was always their weakest output.

Which produces the design recommendation:

Learned position + geometric orientation. A hybrid.

The objectives, honestly

Objective	Verdict	Evidence
1 - Literature review	Met	38 references, limitations stated per paper, gap identified
2 - Person perception	Met	YOLOv8n 99.7% recall vs an 80% target
3 - Group / O-space estimation	NOT MET	11/30 = 36.7% vs a >=70% target
4 - BC vs rule comparison	Completed, with a split result	Position target <0.4 m: met (tuned RF, 0.365 m). Orientation target <20°: not met (best 25.8° after a 360-model search)

More precise than "three of four met." Objective 4's own two numeric targets split - which is the finding, not a caveat.

The closing statement

Every live behavioural experiment in this project was performed in Gazebo simulation. No physical TIAGo experiment and no human-participant study was conducted.

That boundary was set in the proposal from the beginning, and is the reason a new LEAS ethics application was not required. It is a limitation and it is also a design decision, and the documentary should state it plainly rather than let a viewer assume otherwise.

The numbers to show on screen at the end

65 commits · 3 ROS 2 packages · 59 scripts · 24 sessions · 229,678 rows · 462 approach events · 7 features · 56-model tuning search · 360-model systematic search · 22 documented engineering faults · 60 trials · 180 group visits · 30 hand-labelled frames · 93 pages · 38 references · 1 objective honestly failed

What to film / show

A-roll: full beard. This is the reflection/finale, not another bug-fixing episode.

Available video: demo_recordings/YOLO_RUN_VDO.mp4 - use this as the main final-system footage. The project record says a LocateAnything demo clip was still an open item; if you want a detector comparison in the documentary, record that separately and label it as a demonstration capture, not a historical trial unless it is one.

Exact result folder: dataset/processed/results_FINAL_20260824/ with yolo/, locateanything/ and figures/.

Exact figures: fig1_position_accuracy.png, fig2_orientation_accuracy.png, fig3_error_distribution.png, fig4_tradeoff.png, fig5_per_group.png (SVG versions also exist).

Exact analysis file: scripts/approach_accuracy.py; narrative record docs/APPROACH_ACCURACY_RESULTS.md.

Source pages/images: Dissertation pp.73-75 slot-accuracy plots/tables and pp.78-87 research questions/conclusion. Make the position/orientation inversion the final scientific reveal.

Closing screen: 65 commits | 3 ROS 2 packages | ~59 scripts | 24 sessions | 229,678 rows | 462 inferred events | 56-config search | 360-model search | 22 documented faults | 60 final trials | 180 group visits | 30 hand-labelled frames | 93 pages | 38 references.

Accuracy caption: the binary-metric table and slot-accuracy table can refer to different model generations/experimental runs. When showing binary numbers, label the model version and date. The slot-accuracy inversion is the safest headline because it is consistent across the dissertation, repository final files and recomputation described in the project record.

APPENDIX A - AI ASSISTANCE AND AUTHORSHIP MAP

The documentary should be transparent about both research failures and authorship. Git history can show when files changed, but it cannot by itself prove who supplied an idea or a line of reasoning. The map below follows the user's own project record plus the uploaded Phase 4-13 documentary notes.

Stage	Attribution for the documentary
Research direction, supervisor meetings, academic decisions, lab work and all submitted work	Saarunathan Thuviprakash
Phase 1 research scoping / literature exploration	Saarunathan with ChatGPT (Sara) ; supervisor guided the academic direction
Phase 2 proposal refinement / project planning	Saarunathan with ChatGPT assistance , with supervisor feedback and Saarunathan responsible for submission
Phase 3 TIAGo / ROS / Docker learning	Saarunathan with substantial ChatGPT troubleshooting/explanation assistance
Phase 4 early dataset extraction / data understanding	Saarunathan with ChatGPT assistance ; later record reconstructed from code and data
Phases 5-6 early social representation / first BC models	Predominantly Saarunathan with AI assistance ; the uploaded record specifically credits GPT in early pipeline work
Phase 7 Gazebo restaurant world	Saarunathan with extensive ChatGPT (Sara) assistance for environment design, Gazebo/XML edits, actors, waypoints, speeds, assets, collisions, maps and iterative troubleshooting
Phase 8 early ROS integration	Saarunathan with AI assistance ; later reliability fixes span the transition into the Claude-assisted period
Phase 9 detector comparison / LocateAnything debugging	Uploaded project record attributes substantial late-stage assistance to Claude , with Saarunathan running and deciding the experiments
Phase 10 measurement-fault diagnosis / re-scoring	Saarunathan with Claude assistance according to the uploaded record
Phase 11 V2 / sweeps / 360-model search	Saarunathan with Claude assistance according to the uploaded record
Phase 12 O-space tooling / validation	Saarunathan with Claude assistance ; the source record also documents two AI-assisted labelling-workflow bugs that were detected and corrected
Phase 13 slot-accuracy concept	The uploaded record credits the core metric idea to Saarunathan , with implementation/analysis assistance from Claude
Experimental runs, manual clicks, acceptance/rejection decisions	Saarunathan

Recommended public disclosure wording: "I used AI assistants, including ChatGPT and Claude, as research/coding support for literature organisation, troubleshooting, code development, simulation-world construction, analysis and drafting. I performed the experimental runs and manual labelling, discussed academic decisions with my supervisor, and remained responsible for the methodology, validation, interpretation and submitted work."

APPENDIX B - MASTER REPOSITORY / VISUAL-ASSET CHECKLIST

Main ROS 2 runtime nodes

- src/tiago_group_approach/tiago_group_approach/group_perception_node.py
- src/tiago_group_approach/tiago_group_approach/group_approach_baseline_node.py
- src/tiago_group_approach/tiago_group_approach/bc_policy_node.py
- src/tiago_group_approach/tiago_group_approach/mission_node.py
- src/tiago_group_approach/tiago_group_approach/explore_node.py
- src/tiago_group_approach/tiago_group_approach/gt_localisation_node.py
- src/tiago_group_approach/tiago_group_approach/metrics_recorder_node.py

Social-world files

- src/tiago_social_worlds/worlds/restaurant_testing.world
- src/tiago_social_worlds/worlds/restaurant_testing.groundtruth.json
- src/tiago_social_worlds/worlds/restaurant_testing.world.pre_collisions
- src/tiago_social_worlds/worlds/restaurant_final.world
- src/tiago_social_worlds/worlds/restaurant_final.groundtruth.json
- src/tiago_social_worlds/maps/ - generated Nav2 maps
- src/tiago_social_worlds/rviz/ - project RViz layouts

Data / training / analysis scripts visible in the repository

add_gazebo_ros_plugins.py, add_person_collisions.py, approach_accuracy.py, build_approach_pose_dataset.py, build_approach_pose_dataset_v2.py, build_master_session_summary.py, check_dataset_readiness.py, check_gpu.sh, check_sim_setup.sh, cluster_groups.py, drive_test.sh, estimate_approach_points.py, evaluate_approach_pose.py, export_detector_comparison_frames.py, extract_clean_ospace_frames.py, extract_person_detections.py, extract_training_table.py, extract_world_groundtruth.py, finetune_on_lab_pc.py, generate_social_world.py, grid_search_approach_pose.py, inspect_dataset_session.py, install_sim_stack.sh, label_ospace_frames.py, locateanything_service.py, make_nav2_camera_params.py, make_transfer_bundle.sh, merge_ospace_clicks.py, plot_results.py, preflight_check.sh, prepare_lirs_selected_models.py, prepare_ospace_validation.py, record_demo.sh, rescore_sim_results.py, run_everything.sh, run_localisation.sh, run_locateanything_comparison.py, run_nav2_camera.sh, run_overnight.sh, run_pipeline.sh, run_trials.sh, setup_gazebo_env.sh, setup_locateanything.sh, split_dataset.py, start_localisation.sh, summarise_sim_results.py, sweep_dataset_v2.py, sweep_decimation.py, test_locateanything_detector.py, test_person_detector.py, train_approach_pose_mlp.py, train_approach_pose_model.py, train_baseline_model.py, train_evaluate_v2.py, train_social_baseline_model.py, validate_detector_recall.py, validate_ospace_estimate.py, world_to_map.py.

Project documentation files worth using as on-screen evidence

- README.md
- Proposal_Feedback_Action_Plan.md
- Proposal_Gap_Analysis.md
- Introduction_Draft.md
- Literature_Review_Draft.md
- Methodology_Draft.md
- Implementation_Draft.md
- Results_Discussion.md
- Conclusion_Draft.md
- References_Draft.md
- docs/PROJECT_RECORD_POST_REPORT1.md
- docs/V2_RETRAINING_STUDY.md
- docs/APPROACH_ACCURACY_RESULTS.md
- docs/RESULTS_TABLES.md
- docs/RUNBOOK.md
- docs/LAB_PC_SETUP.md
- docs/Section_3_6_Evaluation_Metrics.md

- docs/tiago_setup_progress.md
- TIAGo_Project_Overview.pptx

Final-result visuals

- dataset/processed/results_FINAL_20260824/figures/fig1_position_accuracy.png
- fig2_orientation_accuracy.png
- fig3_error_distribution.png
- fig4_tradeoff.png
- fig5_per_group.png
- SVG versions of all five figures also exist.

Available / missing video and image items

- **Available:** demo_recordings/YOLO_RUN_VD0.mp4 - final YOLO demonstration footage.
- **Record if desired:** a LocateAnything demonstration clip, clearly labelled as a demo capture unless it is an original trial recording.
- **Optional README images identified as unfinished in the source record:**
docs/images/hero_robot_beside_group.png, docs/images/f_formation_diagram.png,
docs/images/simulation_running.png.

Source-document pages that already contain useful visuals

- Proposal p.7 - Gantt/project-plan image.
- Proposal p.8 - risk-analysis table.
- Dissertation p.30 - overall pipeline figure.
- Dissertation p.35 - RViz person-detection screenshot / detector comparison table.
- Dissertation p.43 - ROS 2 system architecture.
- Dissertation pp.44-45 - group table, Gazebo environment and top-down layout.
- Dissertation pp.73-76 - slot-accuracy plots and per-group results.
- Dissertation pp.76-77 - O-space validation table/result.

APPENDIX C - REVERSE SHOOTING CONTINUITY PLAN

The documentary is edited Phase 1 -> 13 but physically filmed Phase 13 -> 1 so beard growth appears chronological.

Shooting order	Story phase	Appearance guidance
1st	Phase 13	Full beard / most exhausted late-project look
2nd	Phase 12	Full beard, only a minimal trim
3rd	Phase 11	Heavy beard
4th	Phase 10	Heavy-to-medium beard
5th	Phase 9	Medium beard
6th	Phase 8	Medium beard
7th	Phase 7	Medium/short beard
8th	Phase 6	Short beard
9th	Phase 5	Short beard / groomed
10th	Phase 4	Light stubble / groomed
11th	Phase 3	Very light stubble
12th	Phase 2	Clean/groomed
13th	Phase 1	Clean-shaven/fresh start

Continuity rule: before every trim, take front/left/right phone photos under the same light. Keep the same glasses, watch and recurring notebook if possible. Let clothing become gradually cleaner/brighter toward Phase 1, but do not make it look like a costume change every episode.

Performance rule: later phases should sound more tired, precise and evidence-driven; earlier phases should sound more speculative and optimistic. Do not give Phase 1 knowledge that was only discovered in Phase 11.

APPENDIX D - OPEN ACCURACY ITEMS BEFORE FINAL EDIT

1. **LocateAnything timing provenance.** Three figures describe different configurations: 25.63 s/frame in an early stored benchmark artifact, 8.40 s/frame in the submitted dissertation benchmark, and roughly 0.4 s/frame after the 24 August model-card fix. Do not use a speed multiplier without naming the configuration.
2. **Binary-metric experiment provenance.** The dissertation's binary table and the repository's later `bc_ft/mlp_ft` run are different experimental/model generations. Caption binary metrics with model version/date. The slot-accuracy headline is consistent and safe for the documentary climax.
3. **LocateAnything final-trial service build.** If the documentary makes a claim about exactly which inference build generated the 24-25 August LocateAnything trials, verify the service version from logs/commit state first.
4. **Physical-robot claim.** Every reported live behavioural evaluation was in Gazebo simulation. Do not imply a physical TIAGo/human-participant experiment occurred.
5. **Ethics claim.** State the project-scope decision exactly: existing anonymised data + simulation, no new participants, and no new LEAS application required for that scope; new physical/human testing would require fresh ethics review.